MTH 201 **Sequences**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Determine if the sequence converges or diverges. If it converges, find the limit.

a) $a_n = \dfrac{\ln\left(n^2\right)}{\ln\left(n+1\right)}$

Let $f\left(x\right) = \dfrac{\ln\left(x^2\right)}{\ln\left(x+1\right)}$.

We calculate $\lim\limits_{x\to\infty} f\left(x\right) = \lim\limits_{x\to\infty} \dfrac{\ln\left(x^2\right)}{\ln\left(x+1\right)} \stackrel{H}{=} \lim\limits_{n\to\infty} \dfrac{\frac{1}{x^2}\cdot 2x}{\frac{1}{x+1}} = \lim\limits_{n\to\infty}\left(\dfrac{2\left(x+1\right)}{x}\right) = 2$.

So, $\lim\limits_{n\to\infty} a_n = \infty$. Therefore, the sequence diverges.

b) $a_n = \dfrac{\left(-1\right)^n n^2}{n^3+1}$

We consider $\left|a_n\right| = \dfrac{n^2}{n^3+1}$

We calculate $\lim\limits_{n\to\infty}\left|a_n\right| = \lim\limits_{n\to\infty} \dfrac{n^2}{n^3+1}\cdot\dfrac{1/n^3}{1/n^3} = \lim\limits_{n\to\infty}\dfrac{\frac{1}{n}}{1+\frac{1}{n^3}} = \dfrac{0}{1+0} = 0$. Therefore, $\lim\limits_{n\to\infty} a_n = 0$, and the sequence converges to 0.

c) $a_n = \cos\left(n\right)$

We note that since the cosine function oscillates between -1 and 1 then $\lim\limits_{n\to\infty}\cos\left(n\right) DNE$ and therefore, the sequence diverges.

d) $a_n = \tan\left(\dfrac{\pi n}{4n+1}\right)$

Since $\lim\limits_{n\to\infty}\dfrac{\pi n}{4n+1} = \lim\limits_{n\to\infty}\dfrac{\pi n}{4n+1}\cdot\dfrac{1/n}{1/n} = \lim\limits_{n\to\infty}\dfrac{\pi}{4+\dfrac{1}{n}} = \dfrac{\pi}{4}$ and the tangent function is continuous at $\dfrac{\pi}{4}$, we have that

$\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty}\tan\left(\dfrac{\pi n}{4n+1}\right) = \tan\left(\lim\limits_{n\to\infty}\dfrac{\pi n}{4n+1}\right) = \tan\left(\dfrac{\pi}{4}\right) = 1$. Thus, the sequence is convergent.

e) $a_n = \dfrac{\sin(2n)}{1+\sqrt{n}}$

Note that $-1 \le \sin(2n) \le 1 \to \dfrac{-1}{1+\sqrt{n}} \le \dfrac{\sin(2n)}{1+\sqrt{n}} \le \dfrac{1}{1+\sqrt{n}}$.

We see that $\lim\limits_{n\to\infty}\dfrac{-1}{1+\sqrt{n}} = 0$ & $\lim\limits_{n\to\infty}\dfrac{1}{1+\sqrt{n}} = 0$.

Therefore, by the Squeeze theorem, $\lim\limits_{n\to\infty}\dfrac{\sin(2n)}{1+\sqrt{n}} = 0$.

f) $a_n = \dfrac{n!}{(2n)!}$

We calculate

$\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty}\dfrac{n!}{(2n)!} = \lim\limits_{n\to\infty}\dfrac{n!}{(2n)(2n-1)\ldots(n+2)(n+1)n!} = \lim\limits_{n\to\infty}\dfrac{1}{(2n)(2n-1)\ldots(n+2)(n+1)} = 0$.

Therefore, the sequence converges to 0.

g) $a_n = \dfrac{-(4^n)+e^{-n}}{2^{2n}}$

For even terms we have that $a_n = \dfrac{4^n + e^{-n}}{2^{2n}}$ and so for these terms

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{4^n + e^{-n}}{2^{2n}} = \lim_{n\to\infty}\left(\frac{2^{2n}}{2^{2n}} + \frac{e^{-n}}{2^{2n}}\right) = \lim_{n\to\infty}(1+0) = 1$$

For odd terms we have that $a_n = \dfrac{-(2^n)+e^n}{4^n}$ and so for these terms

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{-(4^n)+e^{-n}}{2^{2n}} = \lim_{n\to\infty}\left(\frac{-(2^{2n})}{2^{2n}} + \frac{e^{-n}}{2^{2n}}\right) = \lim_{n\to\infty}(-1+0) = -1$$

Since the even and odd terms approach different values this means that $\lim_{n\to\infty} a_n$ *DNE*

2. Find the limit of the sequence $\left\{\sqrt{5}, \sqrt{\sqrt{5}}, \sqrt{\sqrt{\sqrt{5}}}, \ldots\right\}$

The sequence here is clearly decreasing and has a lower bound of 0. Therefore, this sequence has a limit. If we call this limit L and the sequence a_n we have that $\lim_{n\to\infty} a_n = L$.

We define this sequence as $a_n = 5^{\left(\frac{1}{2}\right)^n}$

Therefore, we have $\lim_{n\to\infty} a_n = \lim_{n\to\infty} 5^{\left(\frac{1}{2}\right)^n} = 5^0 = 1$

3. Kyle is trying to determine whether the sequence $a_n = \dfrac{2n-3}{3n+4}$ claims that since the denominator of the fraction will always be larger than the numerator, that the sequence decreases. Show, using all 3 methods discussed in the notes, just how wrong Kyle is.

<u>*Difference of Consecutive Terms*</u>:

$$a_{n+1} - a_n = \frac{2(n+1)-3}{3(n+1)+4} - \frac{2n-3}{3n+4} = \frac{2n-1}{3n+7} - \frac{2n-3}{3n+4} = \frac{(2n-1)(3n+4)-(2n-3)(3n+7)}{(3n+7)(3n+4)}$$

$$= \frac{(6n^2+5n-4)-(6n^2+5n-21)}{(9n^2+12n+21n+28)} = \frac{17}{9n^2+33n+28} > 0$$

Since $a_{n+1} - a_n > 0 \;\rightarrow\; a_{n+1} > a_n$ which implies that the sequence is increasing.

<u>*Ratio of Consecutive Terms:*</u>

$$\frac{a_{n+1}}{a_n} = \frac{\left(\dfrac{2(n+1)-3}{3(n+1)+4}\right)}{\left(\dfrac{2n-3}{3n+4}\right)} = \left(\frac{2n-1}{3n+7}\right)\left(\frac{3n+4}{2n-3}\right) = \frac{6n^2+5n-4}{6n^2+5n-21} = \frac{6n^2+5n-4}{(6n^2+5n-4)-17} > 1$$

Since $\frac{a_{n+1}}{a_n} > 1 \;\rightarrow\; a_{n+1} > a_n$ which implies that the sequence is increasing.

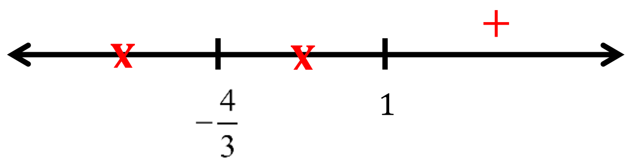
Using Derivatives:

Let $f(x) = \frac{2x-3}{3x+4}$. We calculate that $f'(x) = \frac{(3x+4)(2)-(2x-3)(3)}{(3x+4)^2} = \frac{17}{(3x+4)^2}$.

Clearly there are no values of x that make $f'(x) = 0$.

$f'(x)$ is undefined when $x = -\frac{4}{3}$.

Therefore, the only critical value of $f(x)$ is $x = -\frac{4}{3}$. We test values after this on a number line.



We find that $f'(x) > 0$ for all $x \geq 1$. This implies that $f(x)$ is increasing for $x \geq 1$ and therefore a_n is increasing for $n \geq 1$.

4. Determine if the following sequences are increasing, decreasing, or neither. Also determine if the sequence is bounded.

a) $b_n = \frac{1}{n} - n$

$$b_{n+1} - b_n = \left(\frac{1}{(n+1)} - (n+1)\right) - \left(\frac{1}{n} - n\right) = \frac{1}{n+1} - 1 - \frac{1}{n} = \frac{n - n(n+1) - (n+1)}{n(n+1)}.$$

$$= \frac{n - n^2 - n - 1}{n^2 + n} = \frac{-n^2 - 1}{n^2 + n} = -\frac{(n^2+1)}{n^2+n} < 0$$

Since $b_{n+1} - b_n < 0 \;\rightarrow\; b_{n+1} < b_n$, this implies that b_n decreases.

b) $c_n = (-2)^{n+1}$

We simply look at some of the terms of this sequence.
We can see that $c_n = \{4, -8, 16, \ldots\}$, so clearly the sequence is neither increasing nor decreasing.

5. A sequence a_n with $a_n > 0$ for all n satisfies $\frac{a_{n+1} - a_n}{a_{n+1} + a_n} = \frac{1}{2}$. Suppose $a_1 = 2$.

a) Determine the value of a_5?

$$\frac{a_{n+1} - a_n}{a_{n+1} + a_n} = \frac{1}{2} \quad \to \quad 2\left(a_{n+1} - a_n\right) = a_{n+1} + a_n \quad \to \quad 2a_{n+1} - 2a_n = a_{n+1} + a_n \quad \to \quad a_{n+1} = 3a_n .$$

If $a_1 = 2$ then we have that $a_n = \{2, 6, 18, 54, 162, ...\}$ and therefore $a_5 = 162$

b) Determine if the sequence a_n converges or diverges.

We find that the explicit formula for a_n is $a_n = 2(3)^{n-1} = \frac{2}{3}(3)^n$.

Since $r = 3 > 1$ we know, from the final theorem of section 5.1.1 that this sequence diverges.

6. Suppose the sequence $\{a_n\}$ satisfies $\lim_{n\to\infty}(2n-1)a_n = 16$

a) What is the value of $\lim_{n\to\infty} a_n$?

Notice that $\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left[\frac{(2n-1)a_n}{2n-1}\right] = \lim_{n\to\infty}\left[(2n-1)a_n \cdot \frac{1}{2n-1}\right] = \lim_{n\to\infty}(2n-1)a_n \cdot \lim_{n\to\infty}\frac{1}{(2n-1)} = 16 \cdot 0 = 0$

b) What is the value of $\lim_{n\to\infty} na_n$?

If $\lim_{n\to\infty}(2n-1)a_n = 16$ then

$$\lim_{n\to\infty}(2n-1)a_n = 16 \quad \to \quad \lim_{n\to\infty}(2a_n n - a_n) = 16 \quad \to \quad \lim_{n\to\infty} 2a_n n - \lim_{n\to\infty} a_n = 16$$

$$\to \quad 2\lim_{n\to\infty} a_n n = 16 + \lim_{n\to\infty} a_n \quad \to \quad \lim_{n\to\infty} a_n n = 8 + \frac{1}{2}\lim_{n\to\infty} a_n$$

Then, based on the answer from part (a) we have

$$\lim_{n\to\infty} a_n n = 8 + \frac{1}{2}\lim_{n\to\infty} a_n = 8 + \frac{1}{2}(0) = 8$$

7. Suppose the sequence $\{a_n\}$ and follows the recursion $a_{n+1}^2 = 2a_n + 3$ with $a_1 = 7$. It can be shown that a_n is decreasing. Determine $\lim_{n\to\infty} a_n$.

First note that $a_{n+1} = \sqrt{2a_n + 3}$ and therefore $a_n \geq 0$ for all values of *n*.
By the MST this means that a_n must converge. Suppose $\lim_{n\to\infty} a_n = L$.

We calculate

$$\lim_{n\to\infty} a_{n+1}^2 = \lim_{n\to\infty}(2a_n + 3)$$

$$\left(\lim_{n\to\infty} a_{n+1}\right)^2 = 2\lim_{n\to\infty} a_n + 3$$

Note that $\lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} a_n = L$

This implies that

$$L^2 = 2L + 3$$

$$L^2 - 2L + 3 = 0$$

$$(L-3)(L+1) = 0$$

$$L = 3 \;\; or \;\; L = -1$$

Then, since we know that $a_n \geq 0$ it must be the case that $L = 3$.

So $\lim_{n\to\infty} a_n = 3$